

Deep Learning-Based Face Spoof Detection Using CNN

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This Project is to introduce and present a deep learning approach for face spoof detection using a Convolutional Neural Network (CNN). The system takes in video data by extracting its frames and then applying the CNN model and setting the threshold to classify inputs as real or fake. A controlled dataset with a balanced of real and fake samples was used during testing and when evaluating the testing we used the whole dataset. The model was evaluated using biometric performance metrics such as confusion matrix, False Acceptance Rate (FAR), False Rejection Rate (FRR), Receiver Operating Characteristic (ROC) curve, and Equal Error Rate (EER). While the model received high accuracy results, the tradeoff between false acceptance and usability depended on the decision threshold. The optimal point was determined using the EER, finding the fair balancing point where FAR = FRR.

Additional Key Words and Phrases: face spoof detection, convolutional neural network, biometrics, deep learning, FAR, FRR, EER

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1 Introduction

Facial recognition is widely used in smartphones, airport security, and online identity verification. While facial recognition is convenient, it can be deceived using fake faces such as printed photographs, video replays, and realistic masks. These attacks are known as face spoofing attacks and can allow unauthorized access or identity fraud.

Existing deep learning approaches to face spoof detection have demonstrated strong performance on balanced benchmark datasets. However, real-world deepfake datasets such as FakeAVCeleb exhibit severe class imbalance, where fake samples outnumber real samples by as much as 20:1. This imbalance causes models to bias toward predicting the majority class, significantly degrading their ability to correctly identify genuine faces. Furthermore, the racial and demographic imbalance present in many datasets introduces additional bias, where models may perform better on overrepresented groups and less effectively on underrepresented ones.

This project addresses these two gaps by implementing a fine-tuned MobileNetV2 classifier with targeted class balancing strategies, including forced inclusion of minority class samples, undersampling, and class-weighted loss. The system is evaluated using Equal Error Rate (EER), False Acceptance Rate (FAR), and False Rejection

Rate (FRR) to measure detection fairness rather than raw accuracy alone.

2 Dataset

The dataset used in this project is FakeAVCeleb_v1.2 [1], a large-scale multimodal deepfake dataset designed for audio-visual fake detection research. The dataset contains a total of 21,544 videos spanning multiple racial and demographic groups including Caucasian (American and European), Asian (South and East), and African subjects of both male and female genders.

2.1 Dataset Characteristics

Table 1 summarizes the key characteristics of the FakeAVCeleb dataset.

Table 1. FakeAVCeleb_v1.2 Dataset Characteristics

Property	Value
Total Videos	21,544
Real Videos	1,000
Fake Videos	20,544
Real/Fake Ratio	1:20.5
Frame Resolution	128 × 128 (after preprocessing)
Racial Groups	Caucasian, Asian, African
Genders	Male, Female
Fake Categories	FakeVideo-RealAudio, FakeVideo-FakeAudio

3 Data Engineering

3.1 Train and Test Split

The dataset was divided into training and testing sets using an 80/20 stratified split, ensuring proportional representation of both classes in each set. Table 2 summarizes the resulting split.

Table 2. Train and Test Split

Split	Real	Fake
Train	800	16,435
Test	200	4,109
Total	1,000	20,544

3.2 Class Imbalance

Due to the severe class imbalance of approximately 20:1. This imbalance poses a significant challenge for training, as a naive classifier could achieve approximately 95% accuracy by simply predicting every sample as fake. To address this, targeted balancing strategies were applied during training, which are described in detail in subsection 5.2.

A targeted video selection strategy was applied. All 800 real training videos were forced into the training set to ensure no real samples

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were excluded. To keep training manageable, 3,000 fake videos were randomly sampled from the 16,435 available fake training videos, producing a training pool of 3,800 videos total.

After frame extraction, the training set contained 57,000 frames with a distribution of 45,000 fake and 12,000 real frames. The training set was then undersampled to a 3:1 fake to real ratio using random undersampling, producing a final balanced training set of 48,000 frames containing 36,000 fake and 12,000 real frames.

3.3 Frame Extraction

From each selected video, 15 frames were extracted using random frame sampling rather than uniform sampling. Random sampling was chosen to ensure temporal diversity and prevent the model from seeing redundant frames from the same video segment. This resulted in a total of 57,000 training frames and 14,250 test frames. This also proposed some challenges as we used 15 frames compared to checkpoint 1 where 5 frames was used. This means there were a lot more training to base off of and each epoch would take an average around 6 minutes and 16 epochs which resulted in 96 minutes.

3.4 Face Detection and Cropping

Each extracted frame was processed through a Haar Cascade frontal face detector to crop the face region before resizing. This step was introduced because many video frames contain significant background information that is irrelevant to spoof detection. By cropping to the face region first, the model receives more focused input containing only the facial features relevant to classification. If no face was detected in a frame, the full frame was used as a fallback to avoid discarding data.

3.5 Train and Test Split

The dataset was divided into training and testing sets using an 80/20 stratified split, ensuring proportional representation of both classes in each set. Table 2 summarizes the resulting split.

3.6 Test Set Construction

The full test split contains 200 real and 4,109 fake videos as shown in Table 2. However, to keep evaluation consistent with the training balancing strategy, 750 fake test videos were randomly sampled from the 4,109 available fake test videos. Combined with all 200 real test videos, this produced an evaluation pool of 950 videos and 14,250 test frames after frame extraction at 15 frames per video. The same preprocessing pipeline was applied to test frames as training frames to ensure consistency between training and evaluation.

4 System Architecture

This section presents an overview of the proposed face spoof detection system. Figure 1 illustrates the end-to-end pipeline from raw video input to final classification decision.

4.1 Input Interface

The input to the system consists of video files in MP4 format sourced from the FakeAVCeleb_v1.2 dataset [1]. The dataset contains a total of 21,544 videos across four categories: FakeVideo-FakeAudio, FakeVideo-RealAudio, RealVideo-FakeAudio, and RealVideo-RealAudio.

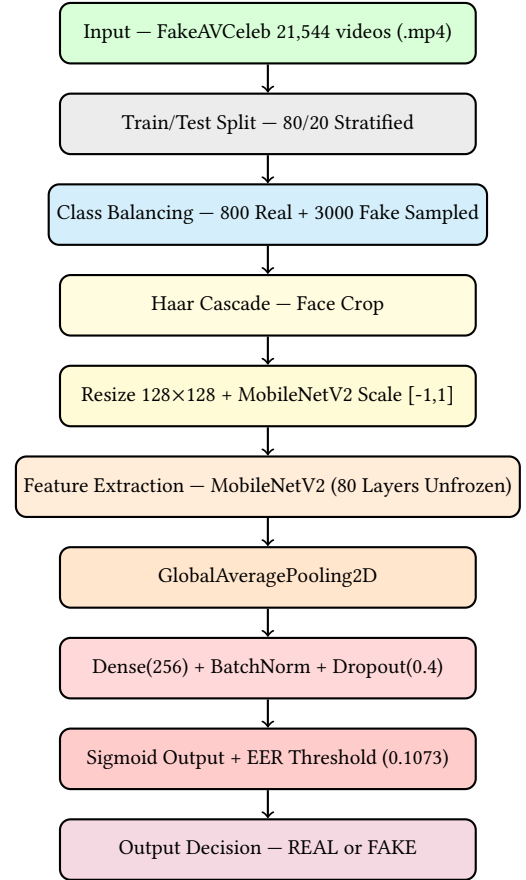


Fig. 1. End-to-end system architecture for face spoof detection using MobileNetV2.

For this project, classification is based solely on the video modality. Videos labeled as FakeVideo are assigned a binary label of 0 (fake) and videos labeled as RealVideo are assigned a binary label of 1 (real).

4.2 Preprocessing Pipeline

Each video undergoes a multi-stage preprocessing pipeline before being fed into the model. First, the dataset is split into training and testing sets using an 80/20 stratified split, ensuring proportional class representation in both sets. To address the severe class imbalance of 20:1, all 800 real training videos are forced into the training set and 3,000 fake videos are randomly sampled, resulting in a total of 3,800 training videos.

From each selected video, 15 frames are extracted using random frame sampling to ensure temporal diversity. Each frame is processed through a Haar Cascade frontal face detector to crop the face region. If no face is detected, the full frame is used as a fallback. The cropped frame is then resized to 128x128 pixels and normalized using MobileNetV2 preprocessing, which scales pixel values to the range [-1, 1].

4.3 Feature Extraction and Template Generation

Feature extraction is performed using MobileNetV2 [8], a lightweight convolutional neural network pretrained on the ImageNet dataset. The top classification layers are removed and replaced with custom layers suited for binary spoof detection. The last 80 layers of MobileNetV2 are set as trainable to allow fine-tuning on the FakeAVCeleb dataset, while the earlier layers retain their pretrained ImageNet weights. A GlobalAveragePooling2D layer is applied after the convolutional base to reduce the spatial feature maps into a single feature vector, which serves as the biometric template for each frame.

4.4 Matcher and Classifier

The extracted feature vector is passed through a classification head consisting of a fully connected Dense layer with 256 units and ReLU activation, followed by Batch Normalization to stabilize training and a Dropout layer with a rate of 0.4 to prevent overfitting. The final output layer consists of a single neuron with a sigmoid activation function, producing a probability score between 0 and 1 representing the likelihood that the input frame is real.

The classification decision is made by applying the Equal Error Rate (EER) threshold of 0.1073, which was determined by finding the point on the ROC curve where the False Acceptance Rate (FAR) and False Rejection Rate (FRR) are equal. Frames with a probability score above this threshold are classified as real, and frames below are classified as fake.

4.5 Database Store

The FakeAVCeleb_v1.2 dataset is stored on disk and organized into category folders by spoofing type, race, and gender. During training, video file paths and their corresponding binary labels are loaded into memory as Python lists and converted into NumPy arrays after frame extraction. The train/test split is stored deterministically using `random_state=42`. The extracted frame arrays `X_train` and `X_test` along with their label arrays `y_train` and `y_test` serve as the in-memory template store used for all training and evaluation operations.

5 Advanced Methodology

This section provides a thorough technical description of the proposed face spoof detection system, including architecture choices, training strategy, and mathematical formalism.

5.1 Architecture Choice

The proposed system is built on MobileNetV2 [8], a lightweight convolutional neural network originally designed for mobile and embedded vision applications. MobileNetV2 was chosen over a custom CNN architecture for several reasons. First, it was pretrained on ImageNet, meaning it already understands low-level visual features such as edges, textures, and facial structures, which are directly relevant to distinguishing real faces from deepfake manipulations. Second, its depthwise separable convolutions make it computationally efficient while maintaining strong feature extraction capability.

The base MobileNetV2 model was loaded with pretrained ImageNet weights and the top classification layers were removed.

The last 80 layers were set as trainable to allow fine-tuning on the FakeAVCeleb dataset, while the earlier layers retained their pretrained weights. This approach, known as transfer learning with partial unfreezing, allows the model to adapt its higher-level feature representations to the specific task of deepfake detection without losing the general visual knowledge learned from ImageNet.

The custom classification head appended on top of MobileNetV2 consists of the following layers:

- **GlobalAveragePooling2D** — reduces the spatial feature maps from the convolutional base into a single 1D feature vector by computing the average of each feature map
- **Dense(256, ReLU)** — a fully connected layer with 256 units and ReLU activation to learn task-specific classification features
- **BatchNormalization** — normalizes the activations of the previous layer to stabilize and accelerate training
- **Dropout(0.4)** — randomly drops 40% of neurons during training to prevent overfitting
- **Dense(1, Sigmoid)** — outputs a single probability score between 0 and 1 representing the likelihood that the input frame is real

5.2 Training Strategy

5.2.1 Class Imbalance Handling. The FakeAVCeleb dataset contains a severe class imbalance of approximately 20:1 fake to real videos. To address this, two strategies were applied. First, all 800 real training videos were forced into the training set with 3,000 randomly sampled fake videos, producing a training pool of 3,800 videos. After frame extraction this resulted in 57,000 training frames. The training set was then undersampled to a 3:1 fake to real ratio using random undersampling, producing a balanced training set of 48,000 frames. Class weights were set to $\{0 : 1.0, 1 : 3.0\}$ to further penalize misclassification of the minority real class during training.

5.2.2 Optimization. The model was compiled using the Adam optimizer [2] with an initial learning rate of 1×10^{-4} . Adam was chosen for its adaptive learning rate properties, which allow it to adjust the effective learning rate for each parameter individually, leading to faster and more stable convergence compared to standard SGD.

The batch size was set to 64, balancing training speed and gradient stability. Training was run for a maximum of 30 epochs with two callbacks to prevent overfitting and improve convergence:

- **EarlyStopping** — monitors validation loss with a patience of 5 epochs and restores the best model weights when training stops
- **ReduceLROnPlateau** — halves the learning rate when validation loss does not improve for 2 consecutive epochs, with a minimum learning rate of 1×10^{-7}

Training converged at epoch 16 with a final validation accuracy of 98.18% and a validation loss of 0.0836. The learning rate was automatically reduced from 1×10^{-4} to 1.25×10^{-5} over the course of training by the ReduceLROnPlateau callback.

5.2.3 Grad-CAM Visualization. To interpret model predictions, Gradient-weighted Class Activation Mapping (Grad-CAM) [9] was applied to visualize which regions of each input frame the model focused on

when making its decision. The heatmaps were generated using the block_16_project convolutional layer of MobileNetV2. Red and yellow regions indicate high model attention, while blue regions indicate low attention. For real videos the model primarily focused on the facial region, while for fake videos the model frequently attended to the chin, jaw, and neck boundary area, which is a known artifact zone in face swap deepfakes where the synthetic face blends imperfectly into the original neck and shoulders.

5.3 Mathematical Formalism

5.3.1 Binary Cross-Entropy Loss. The model is trained using binary cross-entropy loss, which measures the difference between the predicted probability and the true binary label:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (1)$$

where N is the number of training samples, $y_i \in \{0, 1\}$ is the ground truth label, and $\hat{y}_i \in [0, 1]$ is the predicted probability output by the sigmoid layer.

5.3.2 Weighted Loss. To account for the remaining class imbalance after undersampling, class weights were applied during training. The effective weighted loss is:

$$\mathcal{L}_w = -\frac{1}{N} \sum_{i=1}^N w_{y_i} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (2)$$

where w_{y_i} is the class weight for sample i , with $w_0 = 1.0$ for fake samples and $w_1 = 3.0$ for real samples.

5.3.3 Equal Error Rate. The Equal Error Rate (EER) is computed from the ROC curve as the point where the False Acceptance Rate (FAR) equals the False Rejection Rate (FRR):

$$\text{EER} = \text{FAR} = \text{FRR} \quad (3)$$

$$\text{FAR} = \frac{FP}{FP + TN} \quad (4)$$

$$\text{FRR} = \frac{FN}{FN + TP} \quad (5)$$

The EER threshold was determined to be 0.1073, producing an EER of 2.60%, meaning that at this threshold only 2.60% of fake samples are incorrectly accepted and 2.60% of real samples are incorrectly rejected.

6 Exploratory Data Analysis

6.1 Univariate Analysis

The race distribution of the dataset is shown in Figure 2. The dataset contains samples from multiple racial groups, including Caucasian (American and European), Asian (South and East), and African.

Although the dataset includes diverse racial representation, the distribution is not perfectly balanced, with some groups having more samples than others. This imbalance may introduce bias in the model, as it may perform better on more represented groups and less effectively on underrepresented ones.

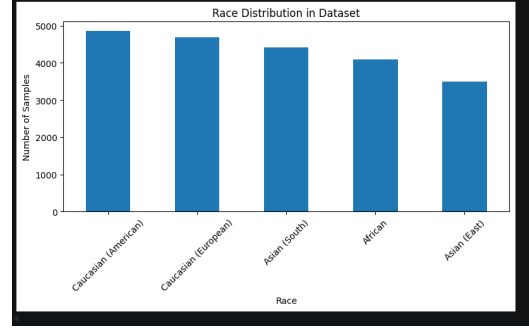


Fig. 2. Distribution of samples across different racial groups in the dataset.

6.2 Bivariate Analysis

Figure 3 shows the distribution of average pixel brightness for real and fake samples. The two distributions exhibit noticeable overlap, indicating that brightness alone is not sufficient to distinguish between real and spoofed inputs.

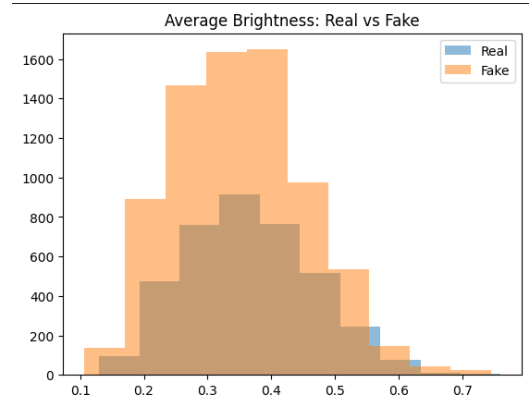


Fig. 3. Distribution of average pixel brightness for real and fake samples.

6.3 Class Balance

The class distribution shows a significant imbalance between real and fake samples. The dataset contains a much larger number of fake samples compared to real samples, as it includes multiple spoofing combinations such as fake video with real audio and fake video with fake audio.

This imbalance can cause bias to the model towards predicting the majority class (fake), potentially reducing its ability to correctly identify real samples.

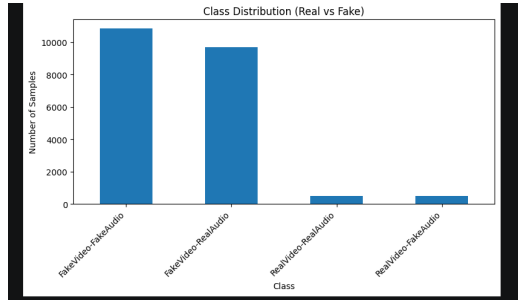


Fig. 4. Class distribution of real and fake samples in the dataset.

7 Related Work

This section reviews existing state-of-the-art methods in face spoof detection and deepfake detection, analyzing how the proposed approach relates to, builds upon, or differs from each method.

7.1 Passive Liveness Detection Using LivenessNet

Priadana and Habibi [6] proposed a passive liveness detection system using a custom deep learning model called LivenessNet. The system operates in two stages: face detection followed by binary classification into real or fake. A key strength of their approach is that it does not require the user to perform any specific facial movements, making it practical for real-world deployment on web and mobile platforms.

Our approach shares the same passive detection philosophy, requiring no user interaction and classifying faces directly from video frames. However, LivenessNet was designed and evaluated primarily on 2D photo and video replay attacks, while our system targets AI-generated deepfake videos from the FakeAVCeleb dataset, which represent a more sophisticated and harder to detect class of spoofing attack. Additionally, our system uses MobileNetV2 with transfer learning rather than a custom architecture trained from scratch, which provides stronger feature representations with less training data.

7.2 Deep Learning for Face Anti-Spoofing: A Survey

Yu et al. [10] provide a comprehensive survey of deep learning techniques for face anti-spoofing, covering texture analysis, depth estimation, pixel-level supervision, and domain adaptation methods. The survey highlights that traditional handcrafted feature methods have become ineffective against modern spoofing techniques and that deep learning has become the dominant approach.

Our work aligns with the survey's findings by adopting a deep learning based approach using a pretrained CNN. However, the survey identifies class imbalance as an underexplored challenge in face anti-spoofing research. Our project directly addresses this gap by implementing forced minority class inclusion, undersampling, and class weighted loss, which are strategies not commonly evaluated in the surveyed methods. The survey also notes that most existing methods focus on 2D attacks, whereas our system targets multimodal deepfake videos.

7.3 Learning Deep Models for Face Anti-Spoofing

Liu et al. [3] proposed improving face anti-spoofing by training CNNs using both binary classification and auxiliary supervision signals such as depth maps and facial surface normals. Their key insight is that using richer supervision signals beyond simple real or fake labels allows the model to learn more discriminative features.

Our approach differs by relying solely on binary classification without auxiliary depth or surface supervision. This makes our system simpler and more practical since depth maps require additional sensors or estimation algorithms. Despite using only binary supervision, our system achieves a significantly lower EER of 2.60% compared to methods that rely on similar binary-only training setups reported in the survey literature. The use of transfer learning from ImageNet partially compensates for the lack of auxiliary supervision by providing rich pretrained feature representations.

7.4 FaceForensics++: Learning to Detect Manipulated Facial Images

Rossler et al. [7] introduced FaceForensics++, a large-scale benchmark dataset for facial manipulation detection containing over 1,000 original videos and manipulated versions generated using four different methods including Deepfakes and FaceSwap. They also proposed using XceptionNet as a backbone for detection, achieving strong results on their benchmark.

Our work is similar in that we both use a pretrained CNN backbone for feature extraction and fine-tune it on deepfake video data. But instead, we use MobileNetV2 instead of XceptionNet, which is more computationally efficient. Our dataset FakeAVCeleb is also more diverse in terms of race and gender representation, which is an important consideration for fairness in biometric systems. A key difference is that FaceForensics++ evaluates on a balanced dataset, whereas our work specifically addresses the challenge of severe class imbalance.

7.5 Multi-Task Learning for Detecting and Segmenting Manipulated Facial Images and Videos

Nguyen et al. [5] proposed a multi-task learning framework that simultaneously detects and segments manipulated regions in facial images and videos. Their approach provides both a binary detection decision and a pixel-level manipulation mask, offering greater interpretability than classification-only methods.

Our approach addresses interpretability differently using Grad-CAM heatmaps, which highlight the regions of each frame that the model focuses on when making its prediction. While Nguyen et al. produce precise pixel-level segmentation masks, our Grad-CAM visualization is simpler to implement and does not require pixel-level ground truth annotations during training. Both approaches are great and they share the same goal of making deepfake detection more interpretable and trustworthy.

7.6 Exploiting Visual Artifacts to Expose Deepfakes and Face Manipulations

Matern et al. [4] proposed detecting deepfakes by exploiting low-level visual artifacts introduced during the face synthesis process, such as eye color inconsistencies, missing facial reflections, and

irregular teeth textures. Their method uses handcrafted visual features rather than deep learning.

Our Grad-CAM analysis is sort of similar to their findings. The heatmaps generated by our model frequently highlight the chin, jaw, and neck boundary areas of fake videos, which are known artifact zones where face swaps blend imperfectly into the original background. This suggests that our MobileNetV2 model implicitly learned to detect similar low-level synthesis artifacts without explicit supervision, making sure that these artifact regions are genuinely discriminative features for deepfake detection.

8 Results and Comparison

Figure 5 shows the training log of the model over 16 epochs. As training progressed, the validation accuracy steadily improved from 80.03% at epoch 1 to 98.27% at epoch 15, while the validation loss decreased from 1.0694 to 0.0802. This consistent improvement indicates that the model was learning meaningful patterns from the data without significant overfitting.

The ReduceLROnPlateau callback automatically halved the learning rate from 0.0001 to 0.00005 after epoch 6 when validation loss plateaued, which resulted in a notable improvement in validation accuracy from 92.89% to 97.99% between epochs 6 and 8. The learning rate continued to reduce throughout training, reaching 0.0000125 by epoch 16.

A patience of 5 was used for early stopping, giving the model sufficient opportunity to recover from temporary fluctuations in validation loss before stopping. Training stopped at epoch 16 with the best weights restored from epoch 11, which had the lowest validation loss of 0.0710.

Epoch 1/16	485s 534m/step	accuracy: 0.9478	loss: 0.1945	val_accuracy: 0.8003	val_loss: 1.0694	learning_rate: 1.0000e-04
750/750						
Epoch 2/16	365s 487m/step	accuracy: 0.9819	loss: 0.0649	val_accuracy: 0.8356	val_loss: 0.7413	learning_rate: 1.0000e-04
750/750						
Epoch 3/16	388s 517m/step	accuracy: 0.9881	loss: 0.0511	val_accuracy: 0.8881	val_loss: 0.5870	learning_rate: 1.0000e-04
750/750						
Epoch 4/16	388s 517m/step	accuracy: 0.9930	loss: 0.0372	val_accuracy: 0.9095	val_loss: 0.5189	learning_rate: 1.0000e-04
750/750						
Epoch 5/16	381s 508m/step	accuracy: 0.9913	loss: 0.0390	val_accuracy: 0.9429	val_loss: 0.2896	learning_rate: 1.0000e-04
750/750						
Epoch 6/16	380s 507m/step	accuracy: 0.9938	loss: 0.0303	val_accuracy: 0.9289	val_loss: 0.3589	learning_rate: 1.0000e-04
750/750						
Epoch 7/16	375s 508m/step	accuracy: 0.9972	loss: 0.0133	val_accuracy: 0.9612	val_loss: 0.1681	learning_rate: 5.0000e-05
750/750						
Epoch 8/16	372s 496m/step	accuracy: 0.9977	loss: 0.0108	val_accuracy: 0.9769	val_loss: 0.0794	learning_rate: 5.0000e-05
750/750						
Epoch 9/16	378s 494m/step	accuracy: 0.9973	loss: 0.0129	val_accuracy: 0.9759	val_loss: 0.1086	learning_rate: 5.0000e-05
750/750						
Epoch 10/16	366s 488m/step	accuracy: 0.9981	loss: 0.0094	val_accuracy: 0.9811	val_loss: 0.0753	learning_rate: 5.0000e-05
750/750						
Epoch 11/16	363s 485m/step	accuracy: 0.9980	loss: 0.0111	val_accuracy: 0.9786	val_loss: 0.0710	learning_rate: 5.0000e-05
750/750						
Epoch 12/16	364s 486m/step	accuracy: 0.9984	loss: 0.0079	val_accuracy: 0.9807	val_loss: 0.0794	learning_rate: 5.0000e-05
750/750						
Epoch 13/16	366s 489m/step	accuracy: 0.9987	loss: 0.0071	val_accuracy: 0.9789	val_loss: 0.0823	learning_rate: 5.0000e-05
750/750						
Epoch 14/16	352s 479m/step	accuracy: 0.9994	loss: 0.0030	val_accuracy: 0.9821	val_loss: 0.0713	learning_rate: 2.5000e-05
750/750						
Epoch 15/16	349s 465m/step	accuracy: 0.9994	loss: 0.0028	val_accuracy: 0.9827	val_loss: 0.0802	learning_rate: 2.5000e-05
750/750						
Epoch 16/16	347s 463m/step	accuracy: 0.9998	loss: 0.0017	val_accuracy: 0.9818	val_loss: 0.0836	learning_rate: 1.2500e-05
750/750						

Fig. 5. Training log showing accuracy, loss, validation accuracy, validation loss, and learning rate over 16 epochs.

8.1 Visual Prediction Analysis

To qualitatively evaluate the model, 20 sample frames were randomly selected from the test set, with 10 real and 10 fake samples chosen to ensure a fair and balanced visual evaluation. This equal split was intentional to verify that the model is genuinely learning to distinguish between real and fake faces rather than defaulting to the majority class.

Figure 6 shows an example of a correctly classified real frame. The model predicted REAL with a confidence score of 0.930, correctly identifying the genuine face.

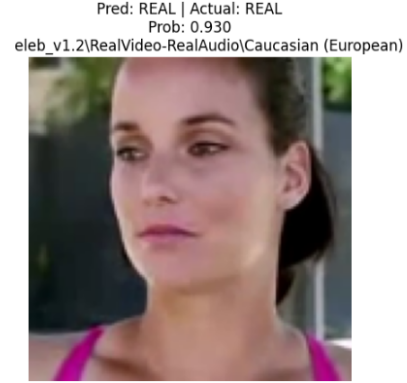


Fig. 6. Example of a correctly classified real frame. Pred: REAL, Actual: REAL, Prob: 0.930.

Figure 7 shows an example of a correctly classified fake frame. The model predicted FAKE with a confidence score of 0.000, indicating extremely high confidence that the frame is a deepfake.



Fig. 7. Example of a correctly classified fake frame. Pred: FAKE, Actual: FAKE, Prob: 0.000.

8.2 Grad-CAM Heatmap Analysis

Figure 8 shows the Grad-CAM heatmaps generated for 5 real samples (top row) and 5 fake samples (bottom row), each from a different video to ensure visual diversity.

For real videos, the model primarily focused on the forehead, cheeks, and upper facial region, indicating that natural skin texture and facial structure are the key features used to confirm genuine faces. For fake videos, the model consistently attended to the chin, jaw, and neck boundary area. This region is a known artifact zone in face swap deepfakes, where the synthetic face blends imperfectly into the original neck and shoulders, producing subtle visual inconsistencies that the model learned to detect.

Notably, the fourth fake sample received a confidence score of 0.100, which is the highest among the fake predictions and closest to the EER threshold of 0.1073. The heatmap for this sample shows

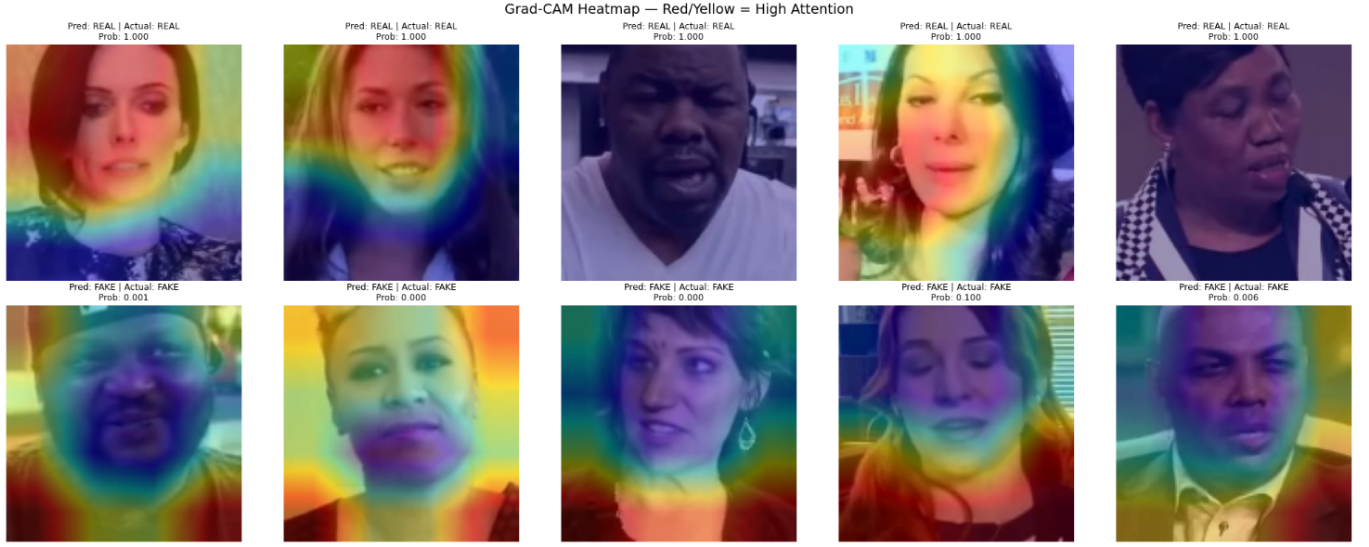


Fig. 8. Grad-CAM heatmaps for 5 real samples (top row) and 5 fake samples (bottom row). Red and yellow regions indicate high model attention, while blue regions indicate low attention.

attention spread across both the face and neck region, suggesting the model was less certain due to a more convincing face swap that produced fewer visible artifacts in the boundary zone.

8.3 Checking Average Correctness from Samples

To further evaluate the model qualitatively, a subset of 20 samples (10 real and 10 fake) was randomly selected from the test dataset. For each sample, the predicted label was compared with the ground truth label to determine whether the prediction was correct or incorrect.

The model correctly classified 20 out of 20 samples, resulting in an accuracy of 100 percent for this subset. This evaluation provides an intuitive understanding of the model's behavior on individual examples.

However, this result is based on a small sample size and does not represent the overall performance of the model. The true performance is better reflected by the full test evaluation using metrics such as accuracy, (FAR), (FRR), and (EER).

8.4 Confusion Matrix

The confusion matrix summarizes the classification performance of the model. It shows the number of correct and incorrect predictions for both real and fake classes. A large number of true negatives indicates that the model is effective at correctly identifying fake samples. However, the presence of false positives and false negatives shows that the model still makes errors in distinguishing between real and fake inputs.

8.5 Performance Metrics

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

$$\text{FAR} = \frac{FP}{FP + TN} \quad (7)$$

$$\text{FRR} = \frac{FN}{FN + TP} \quad (8)$$

8.6 Results

$$\text{Accuracy} = \frac{2920 + 10958}{10958 + 2920 + 292 + 80} = 0.974 \quad (9)$$

$$\text{FAR} = \frac{292}{292 + 10958} = 0.026 \quad (10)$$

$$\text{FRR} = \frac{80}{80 + 2920} = 0.027 \quad (11)$$

8.7 Equal Error Rate (EER)

The Equal Error Rate (EER) is used to evaluate the balance between the (FAR) and the (FRR). This represent the point in which (FAR) and (FARR) are equal or closest to each other.

To determine the Equal Error Rate (EER), the Receiver Operating Characteristic (ROC) curve is generated using the predicted confidence scores. The ROC curve evaluates model performance across different decision thresholds by computing the False Positive Rate (FPR) and True Positive Rate (TPR). The False Negative Rate (FNR), which is equivalent to the False Rejection Rate (FRR), is calculated as:

$$\text{FNR} = 1 - \text{TPR}.$$

The (EER) is then determined by finding the point where the difference between (FPR) and (FNR) is minimized. Here the (EER)

8.8 EER Formula

$$\text{EER} = \text{FAR} = \text{FRR} \quad (12)$$

$$\text{FAR} = \text{FPR} \quad (13)$$

$$\text{FRR} = \text{FNR} \quad (14)$$

8.9 ROC Curve

The Receiver Operating Characteristic (ROC) curve is used to evaluate the performance of the model across different decision thresholds. It plots the False Positive Rate (FPR), also known as the False Acceptance Rate (FAR), against the True Positive Rate (TPR).

Each point on the curve represents a different threshold used to classify the predicted probabilities. The curve illustrates the tradeoff between correctly identifying real samples and incorrectly accepting fake samples.

The Equal Error Rate (EER) is identified on the ROC curve as the point where the False Positive Rate (FPR) and False Negative Rate (FNR) are equal or closest.

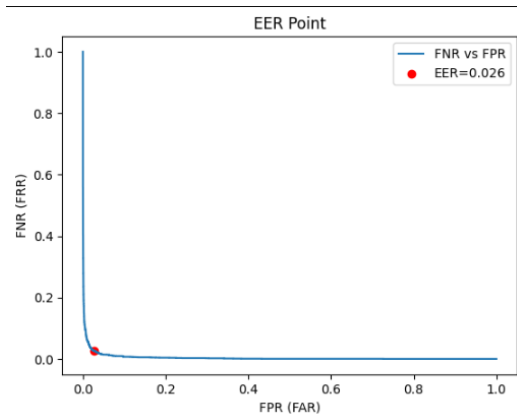


Fig. 9. ROC curve illustrating the relationship between False Positive Rate (FPR) and False Negative Rate (FNR). The highlighted red point represents the Equal Error Rate (EER) of 0.026.

As shown in Figure 9, the ROC curve drops sharply toward the bottom left corner, indicating strong separation between real and fake classes. The EER point is marked at 0.026, meaning that at the optimal threshold of 0.1073, only 2.6% of fake samples are incorrectly accepted and 2.6% of real samples are incorrectly rejected. This is a significant improvement over the checkpoint 1 baseline EER of 22.6%.

9 Error Analysis

Despite achieving strong overall performance with an EER of 2.60% and accuracy of 97.4%, the model exhibits several failure modes that are important to understand for future improvement.

9.0.1 Grad-CAM Interpretability Failure. As shown in Figure 8, the third and fifth real samples in the top row produce flat Grad-CAM heatmaps with no visible red or yellow regions despite being correctly classified with a probability of 1.000. This occurs because the model is so confident in its prediction that the gradient signal

becomes saturated, producing near-zero values across all spatial locations. While the classification is correct, the lack of heatmap coverage means we cannot explain which facial regions the model used to reach its decision. This is a limitation of Grad-CAM specifically on highly confident predictions and represents a failure of interpretability rather than a failure of classification.

9.0.2 Near-Threshold Fake Predictions. The fourth fake sample in the bottom row of Figure 8 received a confidence score of 0.100, which is the closest to the EER threshold of 0.1073 among all fake predictions. This sample nearly fooled the model, and its heatmap shows attention spread across both the face and the neck boundary rather than concentrating on the artifact zone. This suggests that certain face swap manipulations produce fewer visible blending artifacts at the chin and jaw boundary, making them harder for the model to detect confidently.

9.0.3 False Positives and False Negatives. The confusion matrix reveals 292 false positives, where fake frames were incorrectly classified as real, and 80 false negatives, where real frames were incorrectly classified as fake. The false positives represent the most dangerous failure mode in a biometric security context, as they allow spoofed faces to bypass authentication. These errors likely correspond to high quality deepfake videos where the face swap artifacts are minimal and the model cannot detect sufficient visual inconsistencies to classify them as fake.

9.0.4 Low Resolution Limitation. All frames were resized to 128×128 pixels before being fed into the model. This downsampling discards fine-grained visual detail such as subtle skin texture inconsistencies, pore patterns, and micro-expression artifacts that may be visible at higher resolutions. A higher input resolution such as 224×224 or 299×299 may allow the model to detect more subtle deepfake artifacts that are lost during downsampling.

9.0.5 Demographic Bias. The FakeAVCeleb dataset contains an uneven racial distribution, with Caucasian subjects having more samples than African or Asian subjects as shown in the univariate analysis. This imbalance may cause the model to perform better on overrepresented demographic groups and less effectively on underrepresented ones. Future work should evaluate the model separately across racial and gender subgroups to measure detection fairness.

9.0.6 Single Frame Classification. The model classifies each video frame independently without any temporal context from surrounding frames. This means the model cannot detect artifacts that only manifest across multiple frames over time, such as inconsistent blinking patterns, unnatural head movement, or flickering boundaries that appear between frames. Incorporating temporal modeling using recurrent neural networks or 3D convolutions could address this limitation.

10 Conclusion

This paper presented a deep learning based face spoof detection system using a fine-tuned MobileNetV2 convolutional neural network trained on the FakeAVCeleb_v1.2 dataset. The primary challenge addressed was the severe class imbalance of approximately 20:1 fake to real videos, which caused naive classifiers to default to predicting

the majority class. This was addressed through a combination of forced minority class inclusion, random undersampling to a 3:1 ratio, and class weighted loss during training.

The proposed system achieved significant improvements over the checkpoint 1 baseline model across all evaluation metrics as summarized in Table 3.

Table 3. Comparison of Checkpoint 1 Baseline vs Checkpoint 2 Results

Metric	Checkpoint 1	Checkpoint 2
Accuracy	77.4%	97.4%
EER	22.6%	2.6%
FAR	22.6%	2.6%
FRR	22.6%	2.7%

Grad-CAM heatmap analysis revealed that the model learned meaningful and interpretable detection patterns without explicit supervision. For real videos, the model attended to natural facial texture and structure, while for fake videos it consistently focused on the chin, jaw, and neck boundary area where face swap artifacts are known to appear. This finding independently supports existing work on visual artifact based deepfake detection.

Despite strong performance, several limitations remain. The model classifies each frame independently without temporal context, which prevents detection of artifacts that span multiple frames. The 128×128 input resolution discards fine-grained texture detail, and the uneven racial distribution in the dataset may introduce demographic bias in detection performance.

Future work could address these limitations by incorporating temporal modeling using recurrent neural networks or 3D convolutions, increasing input resolution, and evaluating detection fairness across demographic subgroups. Cross-dataset evaluation on unseen deepfake datasets would also provide a stronger measure of generalization beyond FakeAVCeleb.

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